

PROBLEM 1.

$X_1, X_2, \dots, X_n$  are  $n$  independent rvs, where

$X_i \sim \text{Poisson}(\lambda_i \theta)$ , with  $\lambda_i$ 's being given constants.

(i) The likelihood function of  $\theta$  is given by

$$L(\theta | x_1, \dots, x_n) = \prod_{i=1}^n \frac{e^{-\lambda_i \theta} (\lambda_i \theta)^{x_i}}{x_i!}$$

Thus, the log-likelihood function is

$$\begin{aligned} \ell(\theta | x_1, \dots, x_n) &= \log L(\theta | x_1, \dots, x_n) = \sum_{i=1}^n \left\{ -\lambda_i \theta + x_i \log(\lambda_i \theta) - \log x_i! \right\} \\ &= -\theta \sum_{i=1}^n \lambda_i + \sum_{i=1}^n x_i \log(\lambda_i \theta) - \sum_{i=1}^n \log x_i! \end{aligned}$$

Consequently,

$$\frac{\partial \ell(\theta | x_1, \dots, x_n)}{\partial \theta} = -\sum_{i=1}^n \lambda_i + \sum_{i=1}^n \frac{x_i}{\lambda_i \theta} \lambda_i = -\sum_{i=1}^n \lambda_i + \sum_{i=1}^n \frac{x_i}{\theta}$$

Equating the above to zero yields the MLE of  $\theta$  as

$$\hat{\theta}_{MLE} = \frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n \lambda_i} \quad \text{--- (a)}$$

To obtain the least squares estimator of  $\theta$ , we minimize

$$S(\theta) = \sum_{i=1}^n (x_i - \lambda_i \theta)^2$$

$$\frac{\partial S(\theta)}{\partial \theta} = 2 \sum_{i=1}^n (x_i - \lambda_i \theta)(-\lambda_i) = 2 \sum_{i=1}^n \{-\lambda_i x_i + \theta \lambda_i^2\}$$

Equating the above to zero yields the LS estimator of  $\theta$  as

$$\hat{\theta}_{LS} = \frac{\sum_{i=1}^n \lambda_i x_i}{\sum_{i=1}^n \lambda_i^2} \quad \text{--- (b)}$$



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PROBLEM 1 (contd)

$$\begin{aligned} \text{(ii)} \quad \text{From (a), } E(\hat{\theta}_{MLE}) &= \frac{1}{\sum_{i=1}^n \lambda_i} \sum_{i=1}^n E(x_i) \\ &= \frac{1}{\sum_{i=1}^n \lambda_i} \sum_{i=1}^n \theta \lambda_i = \theta. \end{aligned}$$

~~From~~ Thus, the MLE of  $\theta$  is unbiased.

$$\begin{aligned} \text{(iii)} \quad \text{From (b), } E(\hat{\theta}_{LS}) &= \frac{\sum_{i=1}^n \lambda_i E(x_i)}{\sum_{i=1}^n \lambda_i^2} \\ &= \frac{\sum_{i=1}^n \lambda_i (\theta \lambda_i)}{\sum_{i=1}^n \lambda_i^2} = \theta \frac{\sum_{i=1}^n \lambda_i^2}{\sum_{i=1}^n \lambda_i^2} = \theta. \end{aligned}$$

Thus, the MLE and LS estimator of  $\theta$  are both unbiased, and bias is not a good criterion to compare two estimators. Their variances should be compared.



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Problem 2

Let  $X_i, i=1, \dots, n$  be iid samples from  $N(\mu, \sigma^2)$ .  
Let  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ . Prove that  $\bar{X}$  and  $X_i - \bar{X}$  are  
uncorrelated for any  $i$ .

Solution

$$\text{cov}(\bar{X}, X_i - \bar{X})$$

$$= \text{cov}\left(\frac{1}{n} \sum_{i=1}^n X_i, X_i - \bar{X}\right)$$

$$= \text{cov}(X_i, \bar{X}) - \text{cov}(\bar{X}, \bar{X})$$

$$= \text{cov}\left(X_i, \frac{1}{n} \sum_{i=1}^n X_i\right) - \text{var}(\bar{X})$$

$$= \frac{1}{n} \text{cov}\left(X_i, \sum_{i=1}^n X_i\right) - \text{var}\left(\frac{\sum_{i=1}^n X_i}{n}\right)$$

$$= \frac{1}{n} \text{cov}(X_i, X_i) - \frac{1}{n^2} \sum_{i=1}^n \text{var}(X_i)$$

[The first term follows from independence of  $X_i$  and all  $X_j$  for  $j \neq i$ . Second term follows from independence of  $X_1, \dots, X_n$ ]

$$= \frac{1}{n} \text{var}(X_i) - \frac{1}{n^2} \times n \sigma^2$$

$$= \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0$$



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PROBLEM 3.

(i) Let  $X_i$  denote the binary random variable that equals 1 if the flipping of the  $i^{\text{th}}$  coin results in a head ( $i=1, \dots, 10$ ), and 0 otherwise.

Then  $X_i \sim \text{Bern}\left(\frac{i}{10}\right)$ ,  $i=1, \dots, 10$

and  $E(X_i) = \frac{i}{10}$ .

If  $X$  denotes the number of heads when all 10 coins are flipped, then  $X = X_1 + \dots + X_{10}$ , where  $X_1, \dots, X_{10}$  are defined above.

$$\text{Thus, } E(X) = \sum_{i=1}^{10} E(X_i) = \sum_{i=1}^{10} \frac{i}{10} = \frac{1}{10} \sum_{i=1}^{10} i$$

$$= \frac{1}{10} \times \frac{10 \times 11}{2} = \frac{11}{2} = 5.5$$

(ii) Let  $B_i$  denote the event that the  $i^{\text{th}}$  coin is selected ( $i=1, \dots, 10$ ), and let  $A$  denote the event that flipping the chosen coin results in a head. Then we have,

$$P(B_i) = 1/10, \quad i=1, \dots, 10$$

$$P(A|B_i) = i/10, \quad i=1, \dots, 10.$$

We need to find  $P(B_5|A)$ . By application of Baye's theorem,

$$P(B_5|A) = \frac{P(A|B_5)P(B_5)}{\sum_{i=1}^{10} P(A|B_i)P(B_i)} = \frac{\frac{5}{10} \times \frac{1}{10}}{\sum_{i=1}^{10} \frac{i}{10} \times \frac{1}{10}} = \frac{5}{\sum_{i=1}^{10} i} = \frac{5}{\frac{10 \times 11}{2}} = \frac{1}{11}.$$



December 3, 2016Problem 4. $Y$  has a log-normal distribution with density

$$f_Y(y) = \frac{1}{y\sqrt{2\pi\sigma^2}} e^{-\frac{(\log y - \mu)^2}{2\sigma^2}}, \quad y > 0$$

(i) To show that  $Z = \log(Y)$  has a  $\text{Normal}(\mu, \sigma^2)$  distribution, note that the Jacobian of transformation  $y \rightarrow z$  is

$$\left| \frac{\partial y}{\partial z} \right| = \left| \frac{\partial}{\partial z}(e^z) \right| = e^z$$

Thus, the PDF of  $Z$  is

$$f_Z(z) = \frac{1}{e^z \sqrt{2\pi\sigma^2}} e^{-\frac{(z-\mu)^2}{2\sigma^2}} \cdot e^z = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}, \quad -\infty < z < \infty$$

which is the PDF of a  $N(\mu, \sigma^2)$  distribution.

(ii) To calculate the expectation and variance of  $Y$ , we make use of a fundamental result: The moment generating function (MGF) of a  $N(\mu, \sigma^2)$  random variable  $Z$  is given by

$$M_Z(t) = E(e^{tZ}) = e^{t\mu + \frac{1}{2}t^2\sigma^2} \quad \text{for all real } t \quad (a)$$

Now, by part (i),  $Z = \log Y \sim N(\mu, \sigma^2)$ . Hence  $M_Z(t)$  has the form given by (a).

$$\begin{aligned} \text{Now, } E(Y) &= \int_0^\infty y f_Y(y) dy = \int_0^\infty \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\log y - \mu)^2}{2\sigma^2}} dy \\ &= \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(z-\mu)^2} \cdot e^z dz, \quad \text{substituting } z = \log y \end{aligned}$$



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(2) of (3)

Problem 4

(CONTD)

$$= E(e^Z) \quad \text{where } Z \sim N(\mu, \sigma^2)$$

$$= e^{\mu + \frac{1}{2}\sigma^2} \quad \text{from (a), substituting } t=1.$$

To calculate the variance, we obtain

$$E(Y^2) = \int_0^{\infty} y^2 f_Y(y) dy = \int_0^{\infty} y \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\log y - \mu)^2}{2\sigma^2}} dy$$

$$= \int_{-\infty}^{\infty} e^{2z} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(z-\mu)^2}{2\sigma^2}} dz, \quad \text{substituting } z = \log y$$

$$= E(e^{2Z}) \quad \text{where } Z \sim N(\mu, \sigma^2)$$

$$= e^{2\mu + \frac{1}{2} \times 4\sigma^2} \quad (\text{substituting } t=2 \text{ in (a)})$$

$$= e^{2\mu + 2\sigma^2}$$

$$\text{Consequently, } \text{var}(Y) = E(Y^2) - E^2(Y)$$

$$= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1)$$



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Problem 4

(CONT'D)

(iii) To solve this part, we will use the known fact that if  $Z_1, Z_2, \dots, Z_n$  are independent  $N(\mu, \sigma^2)$  variables then  $\tilde{Z} = Z_1 + Z_2 + \dots + Z_n$  has a normal distribution with mean  $n\mu$  and variance  $n\sigma^2$ .

Here,  $Y_1, Y_2, \dots, Y_n \stackrel{iid}{\sim} \text{lognormal}(\mu, \sigma^2)$

Let  $Z_i = \log Y_i, \quad i=1, \dots, n$

Then by part (i),  $Z_1, Z_2, \dots, Z_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

$$\text{Thus, } \tilde{Y} = \prod_{i=1}^n Y_i = \prod_{i=1}^n e^{Z_i} = e^{\sum_{i=1}^n Z_i} = e^{\tilde{Z}}$$

where  $\tilde{Z} = Z_1 + \dots + Z_n \sim N(n\mu, n\sigma^2)$

~~Thus~~ Thus the pdf of  $\tilde{Z}$  is

$$f_{\tilde{Z}}(\tilde{z}) = \frac{1}{\sqrt{2\pi n\sigma^2}} e^{-\frac{1}{2n\sigma^2}(\tilde{z} - n\mu)^2}, \quad -\infty < \tilde{z} < \infty$$

Noting that the Jacobian of transformation  $\tilde{z} \rightarrow \tilde{y}$  is

$$\left| \frac{\partial \tilde{z}}{\partial \tilde{y}} \right| = \left| \frac{\partial}{\partial \tilde{y}} \log \tilde{y} \right| = \frac{1}{|\tilde{y}|}, \quad \text{the PDF of } \tilde{Y} \text{ is}$$

$$f_{\tilde{Y}}(\tilde{y}) = \frac{1}{\tilde{y} \sqrt{2\pi n\sigma^2}} e^{-\frac{1}{2n\sigma^2}(\log \tilde{y} - \mu)^2}, \quad \tilde{y} > 0$$

which by the problem definition is the PDF of a log normal variable with mean  $n\mu$  and variance  $n\sigma^2$ .



# NOTES

①

2015 Dec

2. Joint density of  $X_1$  and  $X_2$ : 
$$\frac{e^{-x_1/\beta}}{\beta^2} x_1 \cdot \frac{e^{-x_2/\beta}}{\beta^2} x_2 = \frac{e^{-(x_1+x_2)/\beta}}{\beta^4} x_1 x_2$$

Hence by the factorization theorem,  $X_1 + X_2$  is sufficient for  $\beta$ .

Also, likelihood function: 
$$L(\beta) = \frac{e^{-(x_1+x_2)/\beta}}{\beta^4} x_1 x_2$$

$$\log L(\beta) = -\frac{(x_1+x_2)}{\beta} - 4 \log \beta + \log(x_1 x_2)$$

$$\frac{d \log L(\beta)}{d\beta} = \frac{x_1+x_2}{\beta^2} - \frac{4}{\beta} = 0 \Rightarrow \beta = \frac{x_1+x_2}{4} = \hat{\beta} \text{ (say)} \quad \left[ \begin{array}{l} \text{note: } \hat{\beta} > 0 \\ \text{with probability 1} \end{array} \right]$$

$$\left. \frac{d^2 \log L(\beta)}{d\beta^2} \right|_{\hat{\beta}} = -\frac{2(x_1+x_2)}{\hat{\beta}^3} + \frac{4}{\hat{\beta}^2} = -\frac{4}{\hat{\beta}^2} < 0$$

$\therefore$  The MLE of  $\beta$  is  $\frac{x_1+x_2}{4}$

2015 Dec

3. (a) For  $k = 0, 1, 2, \dots$  and  $y = k+1, k+2, \dots$

$$P(Y=y|Y>k) = \frac{P(Y=y)}{P(Y>k)} = \frac{q^{y-1} p}{P(Y>k)} \quad (\text{here } q=1-p)$$

$$\therefore E(Y|Y>k) = \frac{1}{P(Y>k)} \sum_{y=k+1}^{\infty} y q^{y-1} p = \frac{1}{P(Y>k)} \sum_{u=1}^{\infty} (u+k) q^{k+u-1} p$$

[Put  $u = y - k$ ]

$$= \frac{q^k}{P(Y>k)} \sum_{u=1}^{\infty} (u+k) q^{u-1} p = \frac{q^k}{P(Y>k)} \left[ \sum_{u=1}^{\infty} u q^{u-1} p + k \sum_{u=1}^{\infty} q^{u-1} p \right]$$

$$= \frac{q^k}{P(Y>k)} \left[ \sum_{u=1}^{\infty} u P(Y=u) + k \sum_{u=1}^{\infty} P(Y=u) \right] = \frac{q^k}{P(Y>k)} [E(Y) + k]$$

$$= \frac{1}{p} + k,$$

because  $E(Y) = \frac{1}{p}$  (given) and  $P(Y>k) = \sum_{u=k+1}^{\infty} P(Y=u)$

$$= \sum_{u=k+1}^{\infty} q^{u-1} p = q^k p (1 + q + q^2 + \dots) = q^k p \frac{1}{1-q} = q^k$$

(b) For  $k = 1, 2, 3, \dots$ ,

$$P(W=k) = P(W \geq k) - P(W \geq k+1)$$

Now,  $P(W \geq k) = P(W_1 \geq k \text{ and } W_2 \geq k \dots \text{ and } W_n \geq k)$

$$= \prod_{i=1}^n P(W_i \geq k) \quad [\text{as } W_1, \dots, W_n \text{ are independent}],$$



## NOTES

and for each  $i$ ,

$$P(W_i \geq k) = \sum_{u=k}^{\infty} P(W_i = u) = \sum_{u=k}^{\infty} q_i^{u-1} p_i \quad [\text{here } q_i = 1 - p_i]$$

$$= q_i^{k-1} p_i (1 + q_i + q_i^2 + \dots) = q_i^{k-1} p_i \frac{1}{1 - q_i} = q_i^{k-1}$$

$$\text{Hence } P(W_i \geq k) = \prod_{i=1}^n q_i^{k-1}. \quad \text{Similarly, } P(W_i \geq k+1) = \prod_{i=1}^n q_i^k$$

$$\therefore P(W_i = k) = \left( \prod_{i=1}^n q_i^{k-1} \right) - \left( \prod_{i=1}^n q_i^k \right), \quad k = 1, 2, \dots$$

(c) If we write  $Z = \max\{W_1, W_2\}$ , then the joint density of  $U$  and  $Z$  is

$$2e^{-(u+3)} \quad 0 \leq u \leq z < \infty$$

Since  $V = Z - U$ , and it is easy to see that the joint density of  $U$  and  $V$  is

$$2e^{-(2u+v)} \quad 0 \leq u < \infty, \quad 0 \leq v < \infty$$

[this can be proved formally by writing, say,  $T = U$ , and then noting that

the transformation  $(U, Z) \rightarrow (T, V)$  has Jacobian unity]

Because  $2e^{-(2u+v)} = (2e^{-2u})(e^{-v})$  and the ranges of  $U$  and  $V$  do not involve each other,  $U$  and  $V$  are independent.

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3. The most powerful test of size  $e^{-1}$  for  $H_0: \theta = 1$  vs  $H_1: \theta = 3$  rejects  $H_0$  if

$$\frac{\frac{1}{3} e^{-2/3}}{e^{-x}} > k \quad \text{for some constant } k$$

$$\text{i.e. if } \frac{1}{3} e^{2x/3} > k, \quad \text{i.e., if } x > k_1 \quad [\text{changing the constant suitably}]$$

Here  $k_1$  is such that

$$e^{-1} = P_{H_0}[X > k_1] = \int_{k_1}^{\infty} e^{-x} dx = e^{-k_1}, \quad \text{i.e., } k_1 = 1$$

Ans: The most powerful test under consideration will reject  $H_0$  if  $x > 1$ .



# NOTES

2015 Dec

1. (a) For  $i=1, \dots, n$ , let  $Y_i$  denote the score of player  $i$ . Then the possible values of  $Y_i$  are 0 and 1, and

$$\begin{aligned} P(Y_i=1) &= \sum_{j=1}^m P(Y_i=1 \mid \text{player } i \text{ selects post } j) P(\text{player } i \text{ selects post } j) \\ &= \sum_{j=1}^m p \left( \frac{1}{m} \right) = p \end{aligned}$$

$\therefore E(Y_i) = p, i=1, \dots, n$   
and  $E(Y_1 + \dots + Y_n) = np$

(b) For  $j=1, \dots, m$ , let  $X_j = 1$  if post  $j$  is hit at the end of the game and 0, otherwise. Then

$$\begin{aligned} P(X_j=0) &= \sum_{k=0}^n P(X_j=0 \mid k \text{ players select post } j) P(k \text{ players select post } j) \\ &= \sum_{k=0}^n (1-p)^k \binom{n}{k} \left( \frac{1}{m} \right)^k \left( 1 - \frac{1}{m} \right)^{n-k} = \sum_{k=0}^n \binom{n}{k} \left( \frac{1-p}{m} \right)^k \left( 1 - \frac{1}{m} \right)^{n-k} \\ &= \left( \frac{1-p}{m} + 1 - \frac{1}{m} \right)^n = \left( 1 - \frac{p}{m} \right)^n \end{aligned}$$

$\therefore P(X_j=1) = 1 - \left( 1 - \frac{p}{m} \right)^n$   
and  $E(X_j) = 1 - \left( 1 - \frac{p}{m} \right)^n$

Hence the expected number of distinct posts hit at the end of the game equals

$$E(X_1 + \dots + X_m) = \sum_{j=1}^m E(X_j) = m \left\{ 1 - \left( 1 - \frac{p}{m} \right)^n \right\}.$$

Alt: For  $i=1, \dots, n$  and  $j=1, \dots, m$ ,

$$\begin{aligned} &P(\text{post } j \text{ is not hit by player } i) \\ &= P(\text{player } i \text{ does not select post } j) + P(\text{player } i \text{ selects post } j \text{ but fails to hit it}) \\ &= 1 - \frac{1}{m} + \frac{1}{m}(1-p) = 1 - \frac{p}{m}. \end{aligned}$$

$$\text{Hence } P(X_j=0) = \prod_{i=1}^n P(\text{post } j \text{ is not hit by player } i) = \left( 1 - \frac{p}{m} \right)^n$$

Now, as before.



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## NOTES

4. (a) Let  $X$  denote the number of steps when the point moves up. If the point starts at  $(0,0)$  then after  $n$  steps its position is  $(n-X, X)$ . Hence we need to calculate  $E\{(n-X)X\}$  where  $X$  has the binomial distribution with parameters  $n$  and  $p$ . Thus the answer is obtained as

$$\begin{aligned} E\{(n-X)X\} &= nE(X) - E(X^2) = nE(X) - \{ \text{var}(X) + (E(X))^2 \} \\ &= n(np) - \{ np(1-p) + n^2 p^2 \} = n^2 p - np(1-p+np) \\ &= np(n-1+p-np) = n(n-1)p(1-p) \end{aligned}$$

(b) Without loss of generality, let  $L=1$  (because we will consider the ratio of the length of the shorter segment to that of the longer segment). Then

$$Y = \frac{\min(X, 1-X)}{\max(X, 1-X)} \quad \text{where } X \text{ has continuous uniform distribution on } [0, 1]$$

For  $0 \leq y \leq 1$ ,

$$\begin{aligned} P(Y \leq y) &= P(Y \leq y \text{ and } X \leq \frac{1}{2}) + P(Y \leq y \text{ and } X > \frac{1}{2}) \\ &= P\left(\frac{X}{1-X} \leq y \text{ and } X \leq \frac{1}{2}\right) + P\left(\frac{1-X}{X} \leq y \text{ and } X > \frac{1}{2}\right) \\ &= P\left(X \leq \frac{y}{1+y} \text{ and } X \leq \frac{1}{2}\right) + P\left(X \geq \frac{1}{1+y} \text{ and } X > \frac{1}{2}\right) \\ &= P\left(X \leq \frac{y}{1+y}\right) + P\left(X \geq \frac{1}{1+y}\right) \quad \left[ \text{as } \frac{1}{1+y} > \frac{1}{2} > \frac{y}{1+y} \right] \\ &= \frac{y}{1+y} + 1 - \frac{1}{1+y} = 1 - \frac{1-y}{1+y} \end{aligned}$$

Also, trivially,  $P(Y \leq 0) = 0$  and  $P(Y \leq 1) = 1$ . Hence, the cdf of  $Y$  is

$$\left. \begin{aligned} F(y) &= 0 & \text{if } y \leq 0 \\ &= 1 - \frac{1-y}{1+y} & \text{if } 0 < y < 1 \\ &= 1 & \text{if } y \geq 1 \end{aligned} \right\}$$